Personality Trait Prediction from Text Using Machine Learning

1. Dataset Description

We used the MBTI dataset from Kaggle. It contains 8,675 forum posts written by users who had previously taken the MBTI test. Each row consists of:

- type: the four-letter personality type (e.g., INFJ, ENTP)
- posts: the text of the user's post (multiple answers separated by ||||)

The posts come from online personality-focused discussions, so the language is natural, unscripted, and suitable for text mining.

Type	Posts
INFJ	'http://www.youtube.com/watch?v=qsXHcwe3krw|||http://41.media.tumblr.com/tumblr_lfouy03PMA1qa1rooo1_500.jpg|||
ENTP	'I'm finding the lack of me in these posts very alarming.|||Sex can be boring if it's in the same position often. For example me
INTP	'Good one _____ https://www.youtube.com/watch?v=fHiGbolFFGw|||Of course, to which I say I know; that's my blessing an
INTJ	'Dear INTP, I enjoyed our conversation the other day. Esoteric gabbing about the nature of the universe and the idea that eve
ENTJ	'You're fired.|||That's another silly misconception. That approaching is logically is going to be the key to unlocking whatever it
INTJ	'18/37 @.@|||Science is not perfect. No scientist claims that it is, or that scientific information will not be revised as we disco
INFJ	'No, I can't draw on my own nails (haha). Those were done by professionals on my nails. And yes, those are all gel. You mean t
INTJ	'I tend to build up a collection of things on my desktop that i use frequently and then move them into a folder called 'Everythi
INFJ	I'm not sure, that's a good question. The distinction between the two is so dependant on perception. To quote Robb Flynn, "Th
INTP	'https://www.youtube.com/watch?v=w8-egj0y8Qs|||I'm in this position where I have to actually let go of the person, due to a
INFJ	'One time my parents were fighting over my dad's affair and my dad pushed my mom. The fall broke her finger. She's pointed
ENFJ	'https://www.youtube.com/watch?v=PLAaiKvHvZs|||51 :o|||I went through a break up some months ago. We were together
INFJ	'Joe santagato - ENTP|||ENFJ or ENTP? I'm not too sure of his type yet|||You know you're not INFJ if heavy Fi doesn't make y
INTJ	'Fair enough, if that's how you want to look at it. Like I stated before, they were incredibly naive in their comments... However
INTP	'Basically this... https://youtu.be/1pH5c1JkhLU|||Can I has Cheezburgr?|||I am very fond of my top hat too. I certainly did not
INTP	'Your comment screams INTJ, bro. Especially the useless part.|||Thanks for the information. Doesn't interfere with anything I'
INFJ	'some of these both excite and calm me: BUTTS bodies brains community gardens camping camping with dogs hiking with c
INFP	'I think we do agree. I personally don't consider myself Alpha, Beta, or Foxtrot (lol at my own joke). People are people. We bot
INFJ	'I fully believe in the power of being a protector, to give a voice to the voiceless. So in that spirit I present this film, and hope i
INFP	'That's normal, it happens also to me. If I am in high mood, I can act like a 478. Depressed, like a 468. Satisfied and relaxed, 451.

2. Problem Statement

We simplified the MBTI framework to focus on two observable dimensions:

- Introvert (I) vs Extrovert (E) – where a person gains energy
- Thinking (T) vs Feeling (F) – how a person makes decisions

The other two axes (Sensing/Intuition, Judging/Perceiving) are harder to detect from short text and were left for future work.

Objective: Build a machine learning model that reads a sentence and predicts both the social style (I/E) and the decision style (T/F) of the writer.

Applications: Social media analytics, customer service tone adjustment, automated personality assessment in psychology research.

3. Data Preparation

From the type column we extracted:

- IE = first letter (I or E)
- TF = third letter (T or F)

The posts column was cleaned by replacing the separator |||| with a space, merging each user's answers into one continuous text.

Text vectorization: We used TfidfVectorizer with:

- max_features=5000
- stop_words='english' (removes words like "the", "and")
- ngram_range=(1,2) (includes single words and bigrams, like "don't like")

TF-IDF gives higher weight to words that are frequent in one post but rare overall, helping the model focus on meaningful words ("quiet", "party") rather than common filler words. Bigrams help capture negation and context.

```python
# Create new columns for each trait and check results
df["IE"] = df["type"].apply(lambda x: x[0])
df["TF"] = df["type"].apply(lambda x: x[2])

df[["type", "IE", "TF"]].head()
```

✓ 0.0s Python

	type	IE	TF
0	INFJ	I	F
1	ENTP	E	T
2	INTP	I	T
3	INTJ	I	T
4	ENTJ	E	T

```python
#Prepare the data by replacing "|||" with space
df["clean_posts"] = df["posts"].str.replace(r"\|\|\|", " ", regex=True)
```

✓ 1.2s Python

4. Methods and Analysis

We performed two classification tasks (I/E and T/F) using the same pipeline.
We present a detailed analysis only for Introvert vs Extrovert
The data was split into 80% training / 20% testing.

We trained and compared two models:

- Naive Bayes (MultinomialNB): A probabilistic model that counts word frequencies per class and multiplies probabilities. Fast and simple baseline.

- Logistic Regression: A linear model that learns a decision boundary separating the two classes. Because the dataset contains more Introverts than Extraverts, we used class_weight='balanced' to give equal importance to both classes.

All models were implemented in Python with scikit-learn.

```python
# Train/Test Split Introvert (I) vs Extrovert (E)
y_IE = df["IE"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y_IE, test_size=0.2, random_state=42
)
```

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```python
# Logistic Regression for I vs E

lr_model_IE = LogisticRegression(max_iter=1000, class_weight="balanced")
lr_model_IE.fit(X_train, y_train)

y_pred_lr = lr_model_IE.predict(X_test)

print(f"Logistic Regression Accuracy (I vs E): {accuracy_score(y_test, y_pred_lr) * 100:.2f}%")
print("\nClassification Report:\n", classification_report(y_test, y_pred_lr))
```

✓ 0.2s Python

```
Logistic Regression Accuracy (I vs E): 84.61%

Classification Report:
               precision    recall  f1-score   support

           E       0.63      0.73      0.68       382
           I       0.92      0.88      0.90      1353

    accuracy                           0.85      1735
   macro avg       0.78      0.80      0.79      1735
weighted avg       0.86      0.85      0.85      1735
```

```python
# Naive Bayes for I vs E
nb_model_IE = MultinomialNB()
nb_model_IE.fit(X_train, y_train)

y_pred_nb = nb_model_IE.predict(X_test)

print(f"Naive Bayes Accuracy (I vs E): {accuracy_score(y_test, y_pred_nb) * 100:.2f}%")
print("\nClassification Report:\n", classification_report(y_test, y_pred_nb))
```

✓ 0.2s Python

```
Naive Bayes Accuracy (I vs E): 78.50%

Classification Report:
               precision    recall  f1-score   support

           E       0.76      0.03      0.07       382
           I       0.79      1.00      0.88      1353

    accuracy                           0.79      1735
   macro avg       0.77      0.52      0.47      1735
weighted avg       0.78      0.79      0.70      1735
```

5. Results and Interpretation

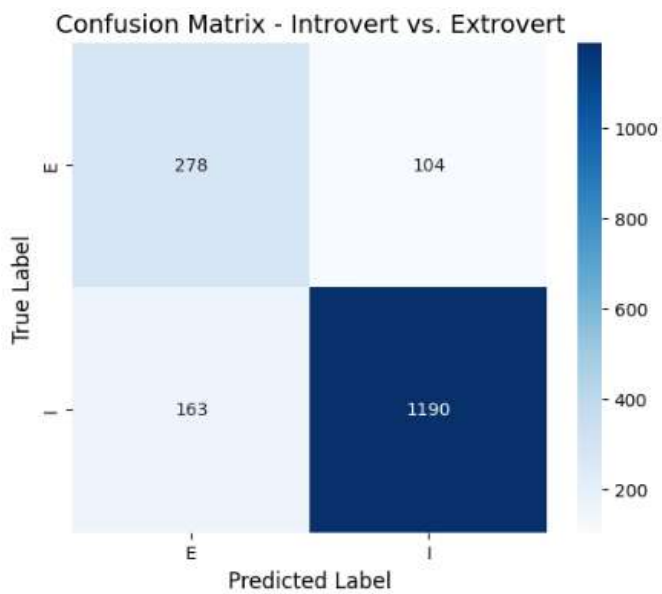
Accuracy comparison:

Model/Task	I vs E	T vs F
Naive Bayes	78.5%	80.1%
Logistic Regression	84.6%	85.3%

Detailed results for I vs E (Logistic Regression):

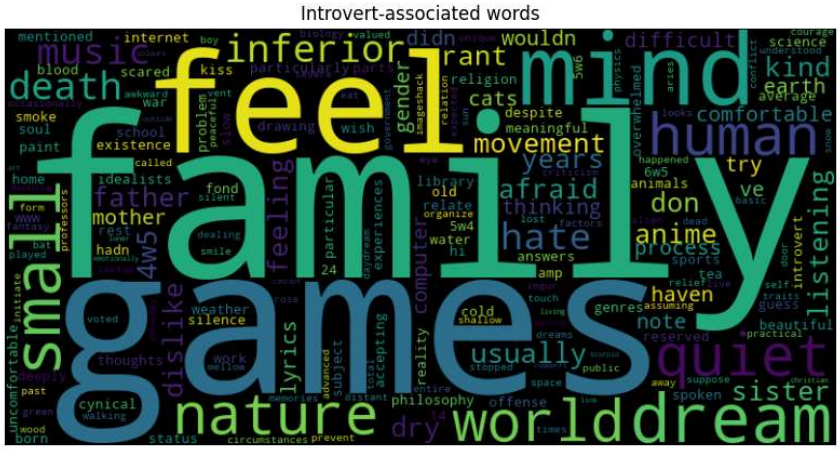
- Accuracy: 84.6%
- Recall for Extrovert: 73% (was only 3% with Naive Bayes)
- Recall for Introvert: 88%

Confusion matrix (I vs E):

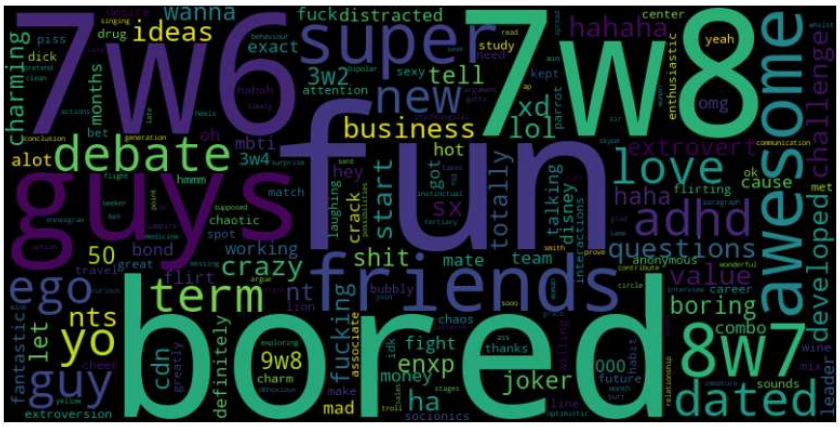


The model still shows a slight bias toward Introvert, but class balancing greatly improved detection of Extroverts.

Word clouds (from Logistic Regression coefficients):


Introvert-associated words
feel
family
games
mind
inferior
nature
world
dream
quiet
human
small
music
death
rant
wouldn't
difficult
kind
comfortable
movement
years
afraid
thinking
hate
don't
anime
process
usually
haven't
note
sister
listening
lyrics
dislike
father
mother
feeling
gender
cats
cold
philosophy
dry
cynical
weather
silence
existence
smoke
paint
school
idealists
hadn't
4w5
5w4
computer
experiences
particular
answers
amp
genres
beautiful
try
ve
animals
science
earth
meaningful
despite
religion
wish
drawing
blood
scared
kiss
internet
mentioned
soul
home
fond
thoughts
work
cynical
spoken
suppose
away
public
stopped
space
offense
memories
distant
circumstances
prevent
status
born
wood
green
past
walking
uncomfortable
initiate
professors
introvert
door
courage
under-sized
conflict
physics
aries
average
looks
happened
6w5
water
h1
organize
relate
lost
factors
old
library
criticism
subject
advanced
accepting
reality
total
trait
rose
living
touch
shallow
dreams
reserved
practical
guess
traits
self
relief
love
tea
sports
assuming
genres
haven't
amp
24
answers
h1
water
5w4
relate
organize

Extrovert-associated words



- Introvert-associated words: quiet, family, feel, mind, dream, hate, afraid
- Extravert-associated words: fun, friends, guys, awesome, debate, love

These patterns align with psychological expectations, validating the model's learned associations

6. Conclusion

We successfully applied data mining to predict personality traits from text.
Key findings:

- Logistic Regression with class balancing clearly outperforms Naive Bayes, especially for imbalanced classes.
- The model achieves 85% accuracy on both I/E and T/F tasks.
- Word clouds reveal interpretable linguistic differences between personality types.

Limitations: The model struggles with negation (e.g., "I don't enjoy being alone" may still be classified as Introvert)

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